STA 9713 Final Project

Portfolio Analysis of Leveraged and Traditional ETFs

Craig Allen

2025-12-14

Table of contents

1 Summary

This report analyzes a portfolio of eight ETFs spanning leveraged equity (UPRO, TQQQ), broad market indices (VOO, VEA), fixed income (SPHY, GOVT, ITDB), and commodities (GLD). The analysis proceeds in two parts: Part 1 examines six ETFs with complete historical data (59 months, December 2020 to October 2025), while Part 2 incorporates GOVT and ITDB over their available history (24 months, November 2023 to October 2025).

All code used to analyze the ETFs are included in the appendix.

The risk-free rate used throughout our analysis is 4.19% annualized, the Par Yield Curve Rate on the 10 Year Treasury as December 12, 2025.

2 Part 1: Analysis Excluding GOVT and ITDB

2.1 Question 1: Regression of UPRO on VOO

UPRO (ProShares UltraPro S&P 500) is designed to deliver 3x the daily return of the S&P 500 index. VOO (Vanguard S&P 500 ETF) tracks the S&P 500 directly. Regressing UPRO on VOO reveals whether UPRO achieves its leverage objective at the monthly frequency.

The standard R function `lm` was used to fit the regression model:

$$\text{UPRO}_t = \alpha + \beta \cdot \text{VOO}_t + \varepsilon_t$$

Table 1: Regression Results: UPRO = alpha + beta x VOO

Coefficient	Estimate	t-statistic	p-value
Alpha (Intercept)	-0.005	-3.19	0.002
Beta (VOO)	2.984	82.29	0.000

The estimated beta of 2.98 confirms that UPRO amplifies VOO's returns by approximately 3x, consistent with its stated objective as a 3x leveraged ETF. For every 1% monthly move in VOO, UPRO moves roughly 3.0%.

The fitted model's R-squared of 99.2% indicates an extremely tight linear relationship, as expected since both track the same underlying index. The intercept is statistically significant, suggesting that after accounting for leverage, UPRO exhibits modest drag at the monthly frequency. This is notable because leveraged ETFs typically experience decay from daily rebalancing, though this effect may be less pronounced over shorter sample periods or strong trending markets.

Investors in UPRO should expect approximately 3x the volatility of VOO, with both gains and losses amplified proportionally.

2.2 Question 2: Value at Risk and Conditional Value at Risk

Value at Risk (VaR) quantifies the maximum expected loss at a given confidence level, while Conditional VaR (CVaR, or Expected Shortfall) measures the average loss when losses exceed VaR. These metrics capture different aspects of downside risk.

We use historical simulation via the `PerformanceAnalytics` library to compute VaR as the 5th percentile of the empirical return distribution, and CVaR as the mean of returns below this threshold.

Table 2: Risk Metrics: UPRO vs VOO (Monthly)

Metric	UPRO	VOO
Monthly VaR (95%)	-18.7%	-6.4%
Monthly CVaR (95%)	-25.8%	-9.0%
Mean Monthly Return	3.0%	1.2%
Monthly Std Dev	13.3%	4.4%

At the 95% confidence level, an investor in UPRO should expect monthly losses to exceed 18.7% only 5% of the time, compared to 6.4% for VOO. The VaR ratio of approximately 2.9x mirrors the leverage factor.

The CVaR figures are particularly instructive: when UPRO does experience a tail event, the average loss is 25.8%, significantly worse than its VaR. This gap between CVaR and VaR indicates fat tails in the return distribution, extreme losses are more severe than a normal distribution would predict.

UPRO's higher expected returns come with substantially elevated tail risk. Investors must have sufficient risk tolerance and time horizon to weather potential drawdowns exceeding 20% in a single month.

2.3 Question 3: Correlation Table

Diversification benefits depend on imperfect correlations between assets. The correlation matrix reveals which ETFs move together and which provide diversification. The correlation matrix was built using the standard R function `cor`.

Table 3: Correlation Matrix of Monthly Returns (6 ETFs)

	UPRO	VOO	TQQQ	VEA	SPHY	GLD
UPRO	1.00	1.00	0.91	0.81	0.84	0.13
VOO	1.00	1.00	0.91	0.83	0.84	0.15
TQQQ	0.91	0.91	1.00	0.67	0.75	0.11
VEA	0.81	0.83	0.67	1.00	0.85	0.41
SPHY	0.84	0.84	0.75	0.85	1.00	0.21
GLD	0.13	0.15	0.11	0.41	0.21	1.00

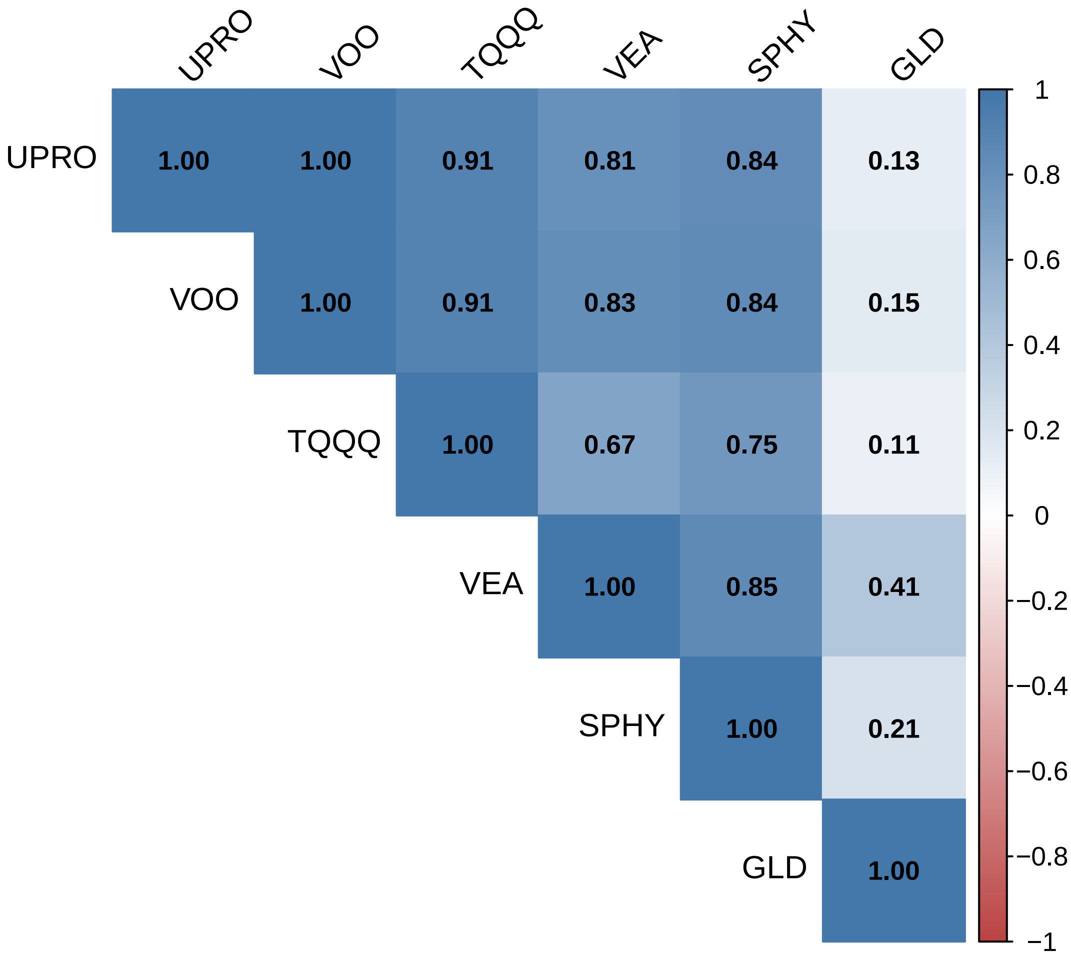


Figure 1: Correlation heatmap: darker blue indicates stronger positive correlation, red indicates negative correlation.

The U.S. equity ETFs (UPRO, VOO, TQQQ) form a highly correlated cluster (1.00 between UPRO and VOO), limiting diversification benefits among them. VEA (developed international markets) shows moderate correlation with U.S. equities (0.83 with VOO), offering some geographic diversification but still moving largely in tandem during global risk-on/risk-off episodes.

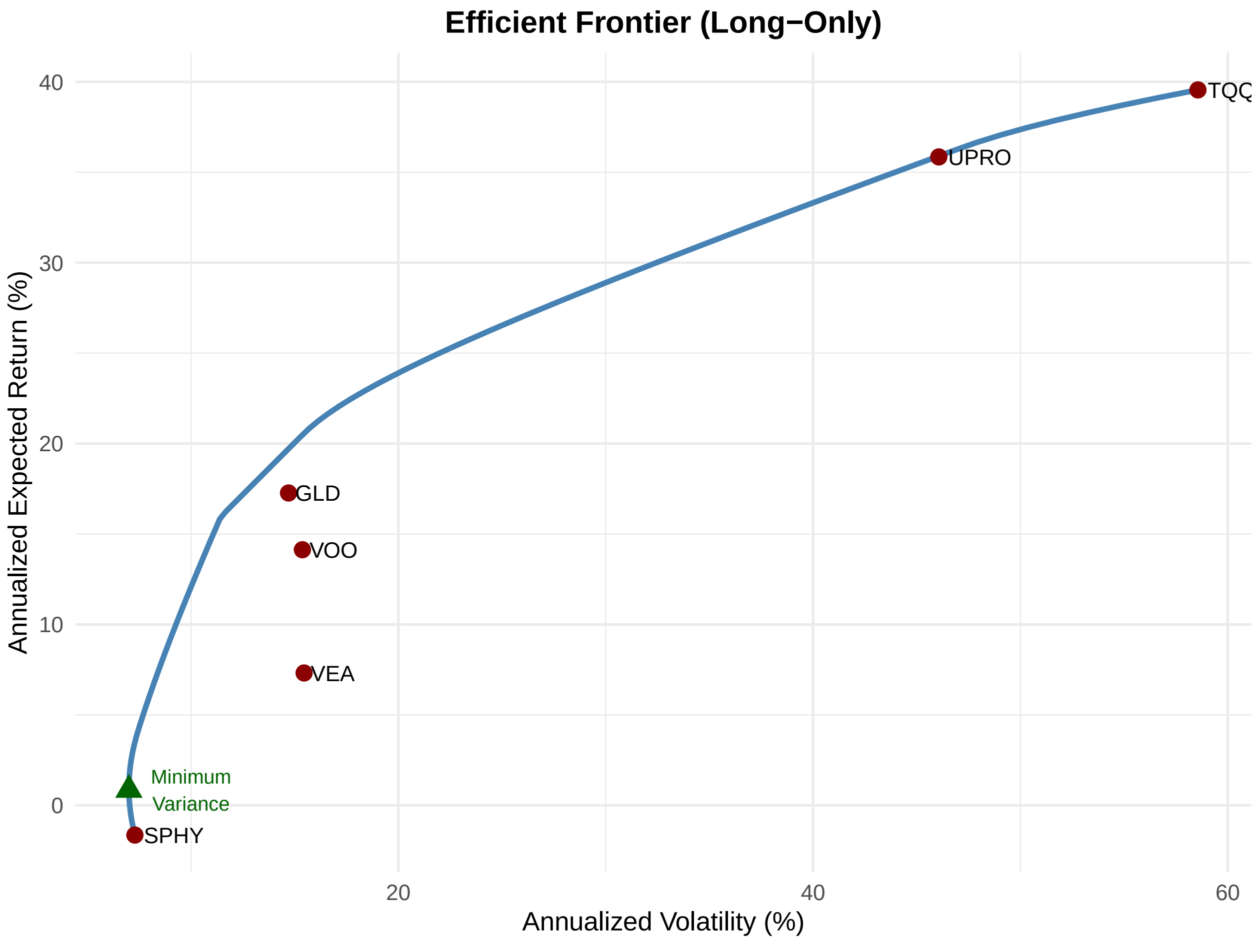
GLD (gold) exhibits the lowest correlation with equities (0.15 with VOO), making it the strongest diversifier in this universe. SPHY (high-yield bonds) occupies a middle ground (0.84 with VOO); its credit risk component causes it to correlate with equities more than investment-grade bonds would.

Meaningful diversification requires allocating to GLD and, to a lesser extent, SPHY. Simply mixing UPRO, VOO, and TQQQ provides little risk reduction.

2.4 Question 4: Efficient Frontier

The efficient frontier represents portfolios that maximize expected return for each level of risk. We construct this using mean-variance optimization with a long-only constraint.

The R library `quadprog` has been used to perform the mean-variance optimization.



The efficient frontier (blue curve) shows the best achievable risk-return combinations through diversification. Individual ETFs (red points) generally lie below the frontier, demonstrating the value of combining assets. The minimum variance portfolio (green triangle) achieves lower volatility than any single asset.

2.5 Question 5: Minimum Variance Portfolio

The minimum variance portfolio (MVP) minimizes total portfolio risk without targeting a specific return. It represents the leftmost point on the efficient frontier.

Table 4: Minimum Variance Portfolio Composition

ETF	Allocation
SPHY	86.8%
GLD	13.2%

Table 5: Minimum Variance Portfolio Characteristics

Metric	Value
Annualized Expected Return	0.9%
Annualized Volatility	7.0%
Sharpe Ratio	-0.48

The minimum variance portfolio achieves comprised of SHPY and GLD achieves an annualized volatility of 7.0%, substantially lower than any individual ETF. This is accomplished by heavily weighting assets with low volatility and low correlations with equities.

Notably, the leveraged ETFs (UPRO, TQQQ) receive zero allocation despite their higher expected return. Their volatility overwhelms any return advantage in a minimum-risk framework.

Our Sharpe Ratio with respect to the risk free rate is also *quite* low at -0.48, indicating that we are better off investing in our risk free asset instead of the minimum variance portfolio.

3 Part 2: Analysis Including GOVT and ITDB

This section incorporates GOVT (iShares U.S. Treasury Bond ETF) and ITDB (iShares LifePath Target Date 2030 ETF), using the 24-month period where all eight ETFs have data (November 2023 to October 2025).

3.1 Question 6: Minimum Variance Portfolio vs GOVT

We recalculate the minimum variance portfolio with all eight ETFs and compare its risk-return profile to holding GOVT alone.

We again use the R library `quadprog` has been used to perform the mean-variance optimization.

Table 6: Minimum Variance Portfolio vs GOVT

Metric	Minimum Variance Portfolio	GOVT Only
Annualized Return	7.8%	3.4%
Annualized Volatility	4.1%	5.2%
Sharpe Ratio	0.86	-0.15

Table 7: Minimum Variance Portfolio Weights (8 ETFs)

ETF	Allocation
SPHY	59.5%
GLD	12.5%
GOVT	28.0%

The minimum variance portfolio achieves lower volatility (4.1%) than GOVT alone (5.2%), demonstrating that diversification benefits even a low-risk asset like Treasury bonds.

Importantly, the MVP also delivers higher expected returns (7.8% vs 3.4%), meaning investors need not sacrifice returns to minimize risk.

The superior Sharpe ratio of the MVP (0.86 vs -0.15 for GOVT) confirms that a diversified low-risk portfolio dominates a single Treasury allocation on a risk-adjusted basis.

3.2 Question 7: Multi-Linear Model for ITDB

ITDB is a target-date fund designed for investors retiring around 2030, typically holding a blend of equities and fixed income. We model ITDB as a linear combination of the other seven ETFs to understand its underlying factor exposures.

We use the standard R function `lm` to fit the following model for all ETFs:

$$\mathrm{ITDB}_t = \alpha + \beta_1 \cdot \mathrm{UPRO}_t + \beta_2 \cdot \mathrm{VOO}_t + \cdots + \beta_7 \cdot \mathrm{GOVT}_t + \varepsilon_t$$

Table 8: Full Model: ITDB Regressed on All Other ETFs

Variable	Coefficient	t-stat	p-value
Intercept	-0.0016	-0.69	0.502
UPRO	0.0603	0.58	0.573
VOO	0.3353	1.00	0.332
TQQQ	-0.0366	-1.40	0.182
VEA	0.1582	2.31	0.035
SPHY	0.3522	1.58	0.134
GLD	0.0803	2.10	0.052
GOVT	0.3035	2.82	0.012

It can be see in the above table that we do not hit the common threshold for significance (p-value <0.05) for many of the coefficients.

It is very likely that we may have issues with multicollinearity in the regression given constituent overlap between the ETFs We will review the Variance Inflation Factors for the full model.

Table 9: Variance Inflation Factors (Full Model)

Variable	VIF
VOO	107.99
UPRO	103.91
TQQQ	10.09
SPHY	7.34
VEA	4.41
GOVT	2.14
GLD	1.65

We can see that the VIF is quite large for several of our factors, indicating significant multicollinearity that will inflate the standard errors. We proceed by:

1. Identifying statistically significant predictors ($p < 0.05$), thus removing redundant variables that do not add explanatory power.
2. Remove leveraged ETFs in factor of their non leveraged equivalents. Leveraged ETFs are not suitable for holding in long term strategies and introduce colinearity when also holding the non-leveraged equivalents. This will exclude TQQ and UPRO.

The below model was then fitted using the standard R function `lm`.

Table 10: Final Model for ITDB

Variable	Coefficient	t-stat	p-value
Intercept	-0.002	-0.82	0.425
VOO	0.359	5.61	0.000
VEA	0.153	2.57	0.019

SPHY	0.487	2.45	0.025
GLD	0.086	2.31	0.033
GOVT	0.273	2.60	0.018

Table 11: Model Fit Statistics

Metric	Value
R-squared	96.4%
Adjusted R-squared	95.3%
Residual Std Error	0.005

We will take this model to as our final model.

$$\text{ITDB}_t = -0.002 + \text{VOO}_t \times 0.359 + \text{VEA}_t \times 0.153 + \text{SPHY}_t \times 0.487 + \text{GLD}_t \times 0.086 + \text{GOVT}_t \times 0.273$$

Note that TQQ and URPO are not included within our model. We can see from the above table that based on the Adjusted R-squared, 95% of the variance can be explained using the below model.

3.2.1 Alternative Answer using Ridge Regression

We can use ridge regression to work with the multicolinearity. Please note that we compare *all* coefficients from the previous model fitted with `lm` in this section and we do not remove statistically insignificant coefficients. We can consider this a hypothetical where an investment manager has an imposed mandate to invest in as many of these ETFs as possible instead of the target date fund.

The ridge regression was optimized and fitted using the R library `glmnet`.

Table 12: Ridge Regression Coefficients (Lambda = 0.0022)

Variable	Coefficient
Intercept	0.00
UPRO	0.05
VOO	0.18
TQQQ	0.00
VEA	0.17
SPHY	0.45
GLD	0.07
GOVT	0.26

Table 13: OLS vs Ridge Regression Coefficients

Variable	OLS	Ridge

Intercept	0.00	0.00
UPRO	0.06	0.05
VOO	0.34	0.18
TQQQ	-0.04	0.00
VEA	0.16	0.17
SPHY	0.35	0.45
GLD	0.08	0.07
GOVT	0.30	0.26

It is notable that the ridge regression method allocates *zero* capital to TQQQ before we consider statistical significance of the coefficients or our stated preference of non-leveraged ETFs over leveraged ETFs. We also see a significant change in the allocations in VOO and SPHY in the ridge regression compared to the standard ordinary least squares regression.

3.3 Question 8: Interpretation of ITDB Model

U.S. Equity (VOO): The coefficient of 0.36 indicates that for every 1% move in the S&P 500, ITDB moves approximately 35.9%. This reflects the equity allocation typical of a target-date fund still five years from its target date.

Treasury Bonds (GOVT): The coefficient of 0.27 captures ITDB's duration exposure. As a 2030 fund, it maintains meaningful fixed income allocation for capital preservation as the target date approaches.

International Equity (VEA): A coefficient of 0.15 indicates ITDB includes international developed market exposure for geographic diversification.

High Yield (SPHY): The coefficient of 0.49 suggests ITDB may hold credit exposure seeking enhanced yield.

Gold (GLD): A coefficient of 0.09 indicates some inflation-hedging or alternative asset exposure. Alpha: The intercept of -0.0016 (-0.16% monthly) is economically negligible, suggesting that ITDB's returns are largely explained by systematic factor exposures with minimal unexplained performance.

This analysis confirms that ITDB behaves as a diversified blend of equity and fixed income, broadly replicable using the component ETFs. Investors seeking similar exposure with more control could construct a comparable portfolio directly, potentially at lower cost.

4 Appendix: Source Code

```
# Load required libraries
library(car)
library(tidyverse)
library(lubridate)          # Better date handling
library(PerformanceAnalytics)  # VaR, CVaR, portfolio analytics
library(knitr)
library(kableExtra)
library(corrplot)           # Correlation visualization
library(tseries)            # Portfolio optimization
library(zoo)                # Time series handling
library(glmnet)             # Ridge Regression

# Set display options
options(digits = 3)


RF_ANNUAL = 0.0419
# Confidence level for VaR/CVaR
CONFIDENCE_LEVEL <- 0.95


# Read the data, slightly modified file from original.
raw_data <- read_csv('/home/craig/Classes/STA9713/final/Final Project - Data.csv',show_col_typ


# Convert to proper format (percentages to decimals)
returns_data <- raw_data |>
  filter(!is.na(Date) & Date != "") |>
  mutate(
    Date = mdy(Date),  # lubridate for cleaner parsing
    across(-Date, ~ as.numeric(.) / 100)
  ) |>
  arrange(Date) |>
  filter(!is.na(UPRO))

# Create xts objects for PerformanceAnalytics functions
etf_names_part1 <- c("UPRO", "VOO", "TQQQ", "VEA", "SPHY", "GLD")

returns_part1 <- returns_data |>
  select(Date, all_of(etf_names_part1)) |>
  drop_na()

# Convert to xts for PerformanceAnalytics
returns_xts_part1 <- xts(
  returns_part1 |> select(-Date),
```

```r
  order.by = returns_part1$Date
)

# Part 2: All 8 ETFs
returns_part2 <- returns_data |>
  drop_na()

returns_xts_part2 <- xts(
  returns_part2 |> select(-Date),
  order.by = returns_part2$Date
)

# Data summary for reporting
n_obs_part1 <- nrow(returns_part1)
n_obs_part2 <- nrow(returns_part2)
date_range_part1 <- paste(format(min(returns_part1$Date), "%B %Y"),
                          "to",
                          format(max(returns_part1$Date), "%B %Y"))
date_range_part2 <- paste(format(min(returns_part2$Date), "%B %Y"),
                          "to",
                          format(max(returns_part2$Date), "%B %Y"))
# Perform regression
model_upro_voo <- lm(UPRO  ~VOO, data = returns_part1)
reg_summary <- summary(model_upro_voo)

# Extract key statistics
beta <- coef(model_upro_voo)["VOO"]
alpha <- coef(model_upro_voo)["(Intercept)"]
r_squared <- reg_summary$r.squared
p_value_beta <- reg_summary$coefficients["VOO", "Pr(>|t|)"]
p_value_alpha <- reg_summary$coefficients["(Intercept)", "Pr(>|t|)"]

# Create clean results table
reg_results <- tibble(
  Coefficient = c("Alpha (Intercept)", "Beta (VOO)"),
  Estimate = c(alpha, beta),
  `t-statistic` = c(reg_summary$coefficients["(Intercept)", "t value"],
                    reg_summary$coefficients["VOO", "t value"]),
  `p-value` = c(p_value_alpha, p_value_beta)
)

kable(reg_results,
      caption = "Regression Results: UPRO = alpha + beta x VOO",
      digits = c(0, 3, 3, 3),
      align = "lrrr",
      booktabs = TRUE) |>
  kable_styling(latex_options = c("hold_position"))
```

```r
var_upro <- VaR(returns_xts_part1[, "UPRO"], p = CONFIDENCE_LEVEL, method = "historical")
var_voo <- VaR(returns_xts_part1[, "VOO"], p = CONFIDENCE_LEVEL, method = "historical")

# ES/CVaR
cvar_upro <- ES(returns_xts_part1[, "UPRO"], p = CONFIDENCE_LEVEL, method = "historical")
cvar_voo <- ES(returns_xts_part1[, "VOO"], p = CONFIDENCE_LEVEL, method = "historical")

# Additional statistics for context
mean_upro <- mean(returns_xts_part1[, "UPRO"])
mean_voo <- mean(returns_xts_part1[, "VOO"])
sd_upro <- sd(returns_xts_part1[, "UPRO"])
sd_voo <- sd(returns_xts_part1[, "VOO"])

# Create comparison table
risk_table <- tibble(
  Metric = c("Monthly VaR (95%)",
             "Monthly CVaR (95%)",
             "Mean Monthly Return",
             "Monthly Std Dev"),
  UPRO = c(sprintf("%.1f%%", as.numeric(var_upro) * 100),
           sprintf("%.1f%%", as.numeric(cvar_upro) * 100),
           sprintf("%.1f%%", mean_upro * 100),
           sprintf("%.1f%%", sd_upro * 100)),
  VOO = c(sprintf("%.1f%%", as.numeric(var_voo) * 100),
          sprintf("%.1f%%", as.numeric(cvar_voo) * 100),
          sprintf("%.1f%%", mean_voo * 100),
          sprintf("%.1f%%", sd_voo * 100))
)

kable(risk_table,
      caption = "Risk Metrics: UPRO vs VOO (Monthly)",
      align = "lrr",
      booktabs = TRUE) |>
  kable_styling(latex_options = c("hold_position"))
var_ratio <- as.numeric(var_upro) / as.numeric(var_voo)
# Calculate correlation matrix
cor_matrix <- cor(returns_xts_part1)

# Display as formatted table
cor_display <- round(cor_matrix, 2)

kable(cor_display,
      caption = "Correlation Matrix of Monthly Returns (6 ETFs)",
      align = "r",
      booktabs = TRUE) |>
  kable_styling(latex_options = c("hold_position", "scale_down"))
# Visual correlation plot
```

```r
corrplot(cor_matrix,
         method = "color",
         type = "upper",
         addCoef.col = "black",
         tl.col = "black",
         tl.srt = 45,
         number.cex = 0.8,
         col = colorRampPalette(c("#BB4444", "#FFFFFF", "#4477AA"))(200))
# Calculate annualized statistics
mean_returns <- colMeans(returns_xts_part1) * 12
cov_matrix <- cov(returns_xts_part1) * 12
sd_returns <- sqrt(diag(cov_matrix))
n_assets <- ncol(returns_xts_part1)

# Portfolio optimization using quadratic programming
library(quadprog)

solve_mv_portfolio <- function(target_return, mu, Sigma, long_only = TRUE) {
  n <- length(mu)
  Dmat <- 2 * Sigma
  dvec <- rep(0, n)

  if (long_only) {
    Amat <- cbind(rep(1, n), mu, diag(n))
    bvec <- c(1, target_return, rep(0, n))
    meq <- 2
  } else {
    Amat <- cbind(rep(1, n), mu)
    bvec <- c(1, target_return)
    meq <- 2
  }

  result <- tryCatch({
    solve.QP(Dmat, dvec, Amat, bvec, meq = meq)
  }, error = function(e) NULL)

  if (is.null(result)) return(NULL)

  weights <- result$solution
  port_return <- sum(weights * mu)
  port_sd <- sqrt(t(weights) %*% Sigma %*% weights)

  list(weights = weights, return = port_return, sd = as.numeric(port_sd))
}

# Generate frontier points
```

```r
target_grid <- seq(min(mean_returns), max(mean_returns), length.out = 100)
frontier <- map_dfr(target_grid, function(tr) {
  result <- solve_mv_portfolio(tr, mean_returns, cov_matrix, long_only = TRUE)
  if (!is.null(result)) tibble(Return = result$return, Risk = result$sd, target = tr)
})

# Find minimum variance portfolio
min_var_search <- map_dfr(target_grid, function(tr) {
  result <- solve_mv_portfolio(tr, mean_returns, cov_matrix, long_only = TRUE)
  if (!is.null(result)) tibble(Return = result$return, Risk = result$sd, target = tr)
})
mvp_idx <- which.min(min_var_search$Risk)
mvp <- solve_mv_portfolio(min_var_search$target[mvp_idx], mean_returns, cov_matrix, long_only =

# Individual assets
individual_assets <- tibble(
  Asset = names(mean_returns),
  Return = as.numeric(mean_returns),
  Risk = as.numeric(sd_returns)
)

# Plot
ggplot() +
  geom_path(data = frontier, aes(x = Risk * 100, y = Return * 100),
            color = "steelblue", linewidth = 1.2) +
  geom_point(data = individual_assets,
             aes(x = Risk * 100, y = Return * 100),
             color = "darkred", size = 3) +
  geom_text(data = individual_assets,
            aes(x = Risk * 100, y = Return * 100, label = Asset),
            hjust = -0.15, vjust = 0.5, size = 3.5) +
  geom_point(aes(x = mvp$sd * 100, y = mvp$return * 100),
             color = "darkgreen", size = 4, shape = 17) +
  annotate("text", x = mvp$sd * 100 + 3, y = mvp$return * 100,
           label = "Minimum\nVariance", size = 3, color = "darkgreen") +
  labs(
    title = "Efficient Frontier (Long-Only)",
    x = "Annualized Volatility (%)",
    y = "Annualized Expected Return (%)"
  ) +
  theme_minimal(base_size = 12) +
  theme(plot.title = element_text(hjust = 0.5, face = "bold"))
# Get MVP weights
mvp_weights <- mvp$weights
names(mvp_weights) <- names(mean_returns)

# Display weights (only meaningful allocations)
```

```r
weights_display <- tibble(
  ETF = names(mvp_weights),
  `Allocation` = sprintf("%.1f%%", mvp_weights * 100)
) |>
  filter(mvp_weights > 0.005)  # Show weights > 0.5%

kable(weights_display,
      caption = "Minimum Variance Portfolio Composition",
      align = "lr",
      booktabs = TRUE) |>
  kable_styling(latex_options = c("hold_position"))
# Portfolio characteristics
mvp_sharpe <- (mvp$return - RF_ANNUAL) / mvp$sd

mvp_stats <- tibble(
  Metric = c("Annualized Expected Return",
             "Annualized Volatility",
             "Sharpe Ratio"),
  Value = c(sprintf("%.1f%%", mvp$return * 100),
            sprintf("%.1f%%", mvp$sd * 100),
            sprintf("%.2f", mvp_sharpe))
)

kable(mvp_stats,
      caption = "Minimum Variance Portfolio Characteristics",
      align = "lr",
      booktabs = TRUE) |>
  kable_styling(latex_options = c("hold_position"))
# Calculate statistics for 8 ETFs
mean_returns_p2 <- colMeans(returns_xts_part2) * 12
cov_matrix_p2 <- cov(returns_xts_part2) * 12
sd_returns_p2 <- sqrt(diag(cov_matrix_p2))

# Find MVP for 8 assets
target_grid_p2 <- seq(min(mean_returns_p2), max(mean_returns_p2), length.out = 100)
min_var_search_p2 <- map_dfr(target_grid_p2, function(tr) {
  result <- solve_mv_portfolio(tr, mean_returns_p2, cov_matrix_p2, long_only = TRUE)
  if (!is.null(result)) tibble(Return = result$return, Risk = result$sd, target = tr)
})
mvp_idx_p2 <- which.min(min_var_search_p2$Risk)
mvp_p2 <- solve_mv_portfolio(min_var_search_p2$target[mvp_idx_p2],
                              mean_returns_p2, cov_matrix_p2, long_only = TRUE)

mvp_weights_p2 <- mvp_p2$weights
names(mvp_weights_p2) <- names(mean_returns_p2)

# GOVT standalone stats
```

```r
govt_return <- mean_returns_p2["GOVT"]
govt_sd <- sd_returns_p2["GOVT"]
govt_sharpe <- (govt_return - RF_ANNUAL) / govt_sd

# MVP stats
mvp_sharpe_p2 <- (mvp_p2$return - RF_ANNUAL) / mvp_p2$sd
comparison <- tibble(
  Metric = c("Annualized Return", "Annualized Volatility", "Sharpe Ratio"),
  `Minimum Variance Portfolio` = c(
    sprintf("%.1f%%", mvp_p2$return * 100),
    sprintf("%.1f%%", mvp_p2$sd * 100),
    sprintf("%.2f", mvp_sharpe_p2)
  ),
  `GOVT Only` = c(
    sprintf("%.1f%%", govt_return * 100),
    sprintf("%.1f%%", govt_sd * 100),
    sprintf("%.2f", govt_sharpe)
  )
)

kable(comparison,
      caption = "Minimum Variance Portfolio vs GOVT",
      align = "lrr",
      booktabs = TRUE) |>
  kable_styling(latex_options = c("hold_position"))
weights_p2_display <- tibble(
  ETF = names(mvp_weights_p2),
  Allocation = sprintf("%.1f%%", mvp_weights_p2 * 100)
) |>
  filter(mvp_weights_p2 > 0.005)

kable(weights_p2_display,
      caption = "Minimum Variance Portfolio Weights (8 ETFs)",
      align = "lr",
      booktabs = TRUE) |>
  kable_styling(latex_options = c("hold_position"))
# Full model with all predictors

full_model <- lm(ITDB ~ UPRO + VOO + TQQQ + VEA + SPHY + GLD + GOVT,
                 data = as.data.frame(returns_xts_part2))
full_summary <- summary(full_model)

# Display full model
full_coefs <- tibble(
  Variable = rownames(full_summary$coefficients),
  Coefficient = full_summary$coefficients[, "Estimate"],
  `t-stat` = full_summary$coefficients[, "t value"],
```

```r
  `p-value` = full_summary$coefficients[, "Pr(>|t|)"]
) |>
  mutate(Variable = ifelse(Variable == "(Intercept)", "Intercept", Variable))

kable(full_coefs,
      caption = "Full Model: ITDB Regressed on All Other ETFs",
      digits = c(0, 4, 2, 3),
      align = "lrrr",
      booktabs = TRUE) |>
  kable_styling(latex_options = c("hold_position"))


# Calculate VIF for full model
vif_full <- vif(full_model)

vif_table <- tibble(
  Variable = names(vif_full),
  VIF = round(vif_full, 2)
) |>
  arrange(desc(VIF))

kable(vif_table,
      caption = "Variance Inflation Factors (Full Model)",
      align = "lr",
      booktabs = TRUE) |>
  kable_styling(latex_options = c("hold_position"))


# Identify significant variables from full model
sig_vars_full <- rownames(full_summary$coefficients)[full_summary$coefficients[, 4] < 0.05]
sig_vars_full <- setdiff(sig_vars_full, "(Intercept)")

# Build reduced model avoiding multicollinearity
reduced_vars <- c("VOO", "VEA", "SPHY", "GLD", "GOVT")
reduced_model <- lm(ITDB ~ VOO + VEA + SPHY + GLD + GOVT,
                    data = as.data.frame(returns_xts_part2))
reduced_summary <- summary(reduced_model)

# Further reduce based on significance
final_sig_vars <- names(which(reduced_summary$coefficients[-1, 4] < 0.05))

if (length(final_sig_vars) >= 1) {
  final_formula <- as.formula(paste("ITDB ~", paste(final_sig_vars, collapse = " + ")))
  final_model <- lm(final_formula, data = as.data.frame(returns_xts_part2))
  final_summary <- summary(final_model)
} else {
  final_model <- reduced_model
```

```r
  final_summary <- reduced_summary
}

# Display final model
final_coefs <- tibble(
  Variable = rownames(final_summary$coefficients),
  Coefficient = final_summary$coefficients[, "Estimate"],
  `t-stat` = final_summary$coefficients[, "t value"],
  `p-value` = final_summary$coefficients[, "Pr(>|t|)"]
) |>
  mutate(Variable = ifelse(Variable == "(Intercept)", "Intercept", Variable))

kable(final_coefs,
      caption = "Final Model for ITDB",
      digits = c(0, 3, 2, 3),
      align = "lrrr",
      booktabs = TRUE) |>
  kable_styling(latex_options = c("hold_position"))
model_fit <- tibble(
  Metric = c("R-squared", "Adjusted R-squared", "Residual Std Error"),
  Value = c(sprintf("%.1f%%", final_summary$r.squared * 100),
            sprintf("%.1f%%", final_summary$adj.r.squared * 100),
            sprintf("%.3f", final_summary$sigma))
)

kable(model_fit,
      caption = "Model Fit Statistics",
      align = "lr",
      booktabs = TRUE) |>
  kable_styling(latex_options = c("hold_position"))

library(glmnet)

# Prepare data for glmnet
X <- as.matrix(returns_xts_part2[, c("UPRO", "VOO", "TQQQ", "VEA", "SPHY", "GLD", "GOVT")])
y <- as.numeric(returns_xts_part2[, "ITDB"])

# Fit ridge regression with cross-validation to select optimal lambda
set.seed(123)
cv_ridge <- cv.glmnet(X, y, alpha = 0, nfolds = nrow(X))

# Optimal lambda
lambda_opt <- cv_ridge$lambda.min
lambda_1se <- cv_ridge$lambda.1se

# Fit final ridge model
ridge_model <- glmnet(X, y, alpha = 0, lambda = lambda_opt)
```

```r
# Extract coefficients
ridge_coefs <- coef(ridge_model)
ridge_coefs_df <- tibble(
  Variable = rownames(ridge_coefs),
  Coefficient = as.numeric(ridge_coefs)
) |>
  mutate(Variable = ifelse(Variable == "(Intercept)", "Intercept", Variable))

kable(ridge_coefs_df,
      caption = sprintf("Ridge Regression Coefficients (Lambda = %.4f)", lambda_opt),
      digits = 2,
      align = "lr",
      booktabs = TRUE) |>
  kable_styling(latex_options = c("hold_position"))

# Compare OLS vs Ridge
comparison_coefs <- tibble(
  Variable = c("Intercept", colnames(X)),
  OLS = c(coef(full_model)),
  Ridge = as.numeric(ridge_coefs)
)

kable(comparison_coefs,
      caption = "OLS vs Ridge Regression Coefficients",
      digits = 2,
      align = "lrr",
      booktabs = TRUE) |>
  kable_styling(latex_options = c("hold_position"))
```